

# A small supervision budget barely improves weak-to-strong generalization

On GPT-2 / BoolQ, mixing a ground-truth budget into the weak labels helps only modestly — and *where* **and** *how* it is spent show little effect.

A consistent account: the student largely reproduces the teacher's errors, and ground truth corrects mainly the examples it directly labels — so the effective lever is *volume*, not placement or combination.

GPT-2 family · BoolQ + SciQ · 3 seeds · paired per-(pair,seed) contrasts · pre-registered

## Scope & setup

---

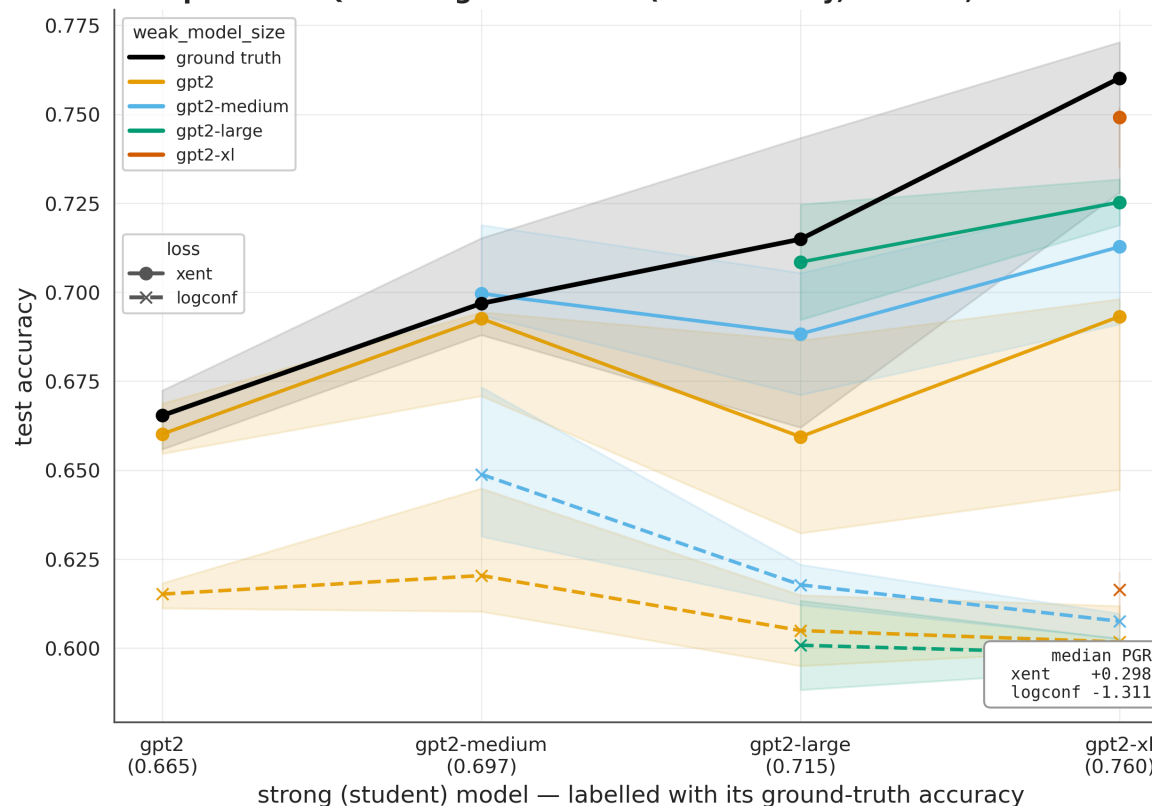
- **Setting:** weak-to-strong generalization — a strong *student* is trained on a weak *teacher's* labels.
- **Extension:** a *supervision budget* — mix a fraction of ground-truth ("strong") labels into the weak labels, and sweep that fraction.
- **Models: GPT-2 family only** — gpt2 / medium / large / xl, with within-family student-teacher pairs.
- **Tasks: BoolQ** (required) + **SciQ** (cross-task check).
- **Readout:** median **PGR** across the model sweep (raw accuracy primary; PGR secondary).

3 seeds · paired per-(pair,seed) contrasts · pre-registered · gpt2-large (seed 1) excluded by rule

## GPT-2 sweep at 25% ground truth

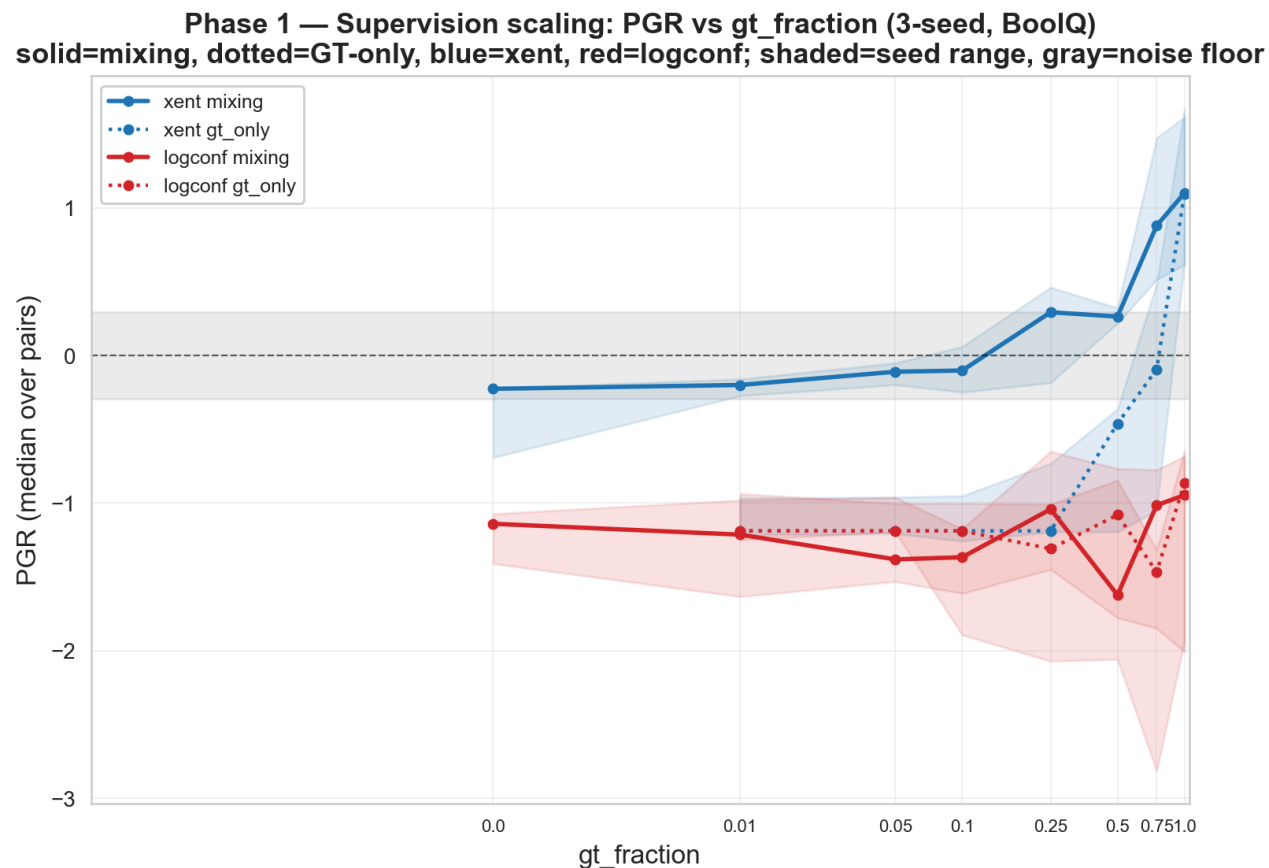
- GPT-2 family, BoolQ, **25% ground truth mixed into the weak labels** — standard sweep format.
- **Median PGR (xent) = +0.30** across the sweep; logconf (dashed) sits well below xent.
- The 0% baseline is **-0.27**; 25% GT moves the median to **+0.30** — a small positive.

W2SG sweep — BOOLQ + 25% ground truth (GPT-2 family, 3 seeds, band = seed range)



## How *much*? — back-loaded, and saturating by ~75%

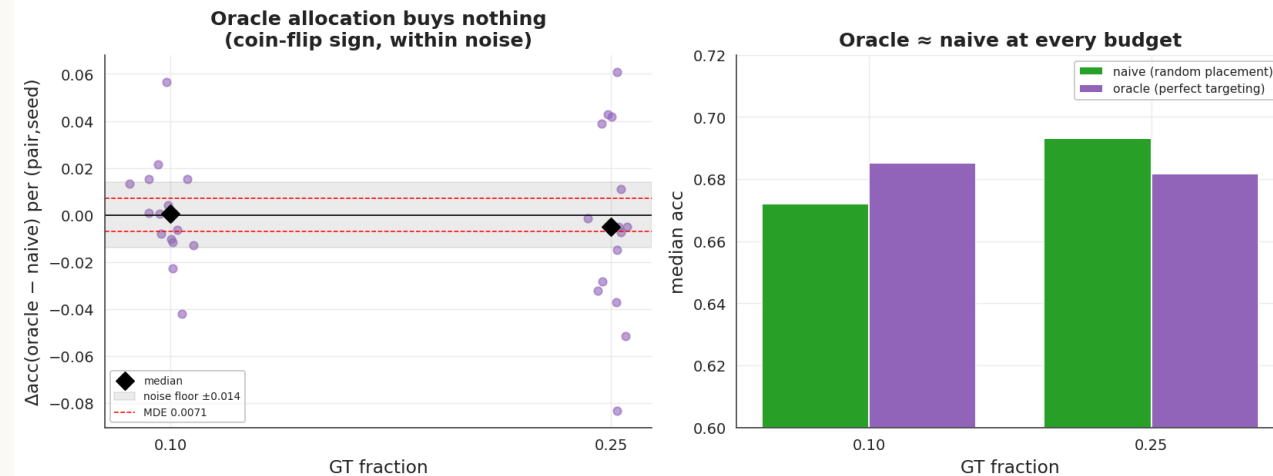
- Up to **10% GT**, the gain over the 0% baseline stays within the **0.014 noise floor**.
- Median xent PGR is **back-loaded**:  $-0.22 \rightarrow +0.30$  (0.25)  $\rightarrow +0.28$  (0.50)  $\rightarrow +0.90$  (0.75)  $\rightarrow +1.04$  (1.0).
- The 0.75 point: the **0.50 $\rightarrow$ 0.75 step is the largest**, and **0.75 $\rightarrow$ 1.0 is within noise** — the gain is essentially complete by ~75%.
- A pre-registered concave-knee-at-25% prediction was **refuted**, and retracted after the multi-seed data.



## Where it's spent — within noise (allocation null)

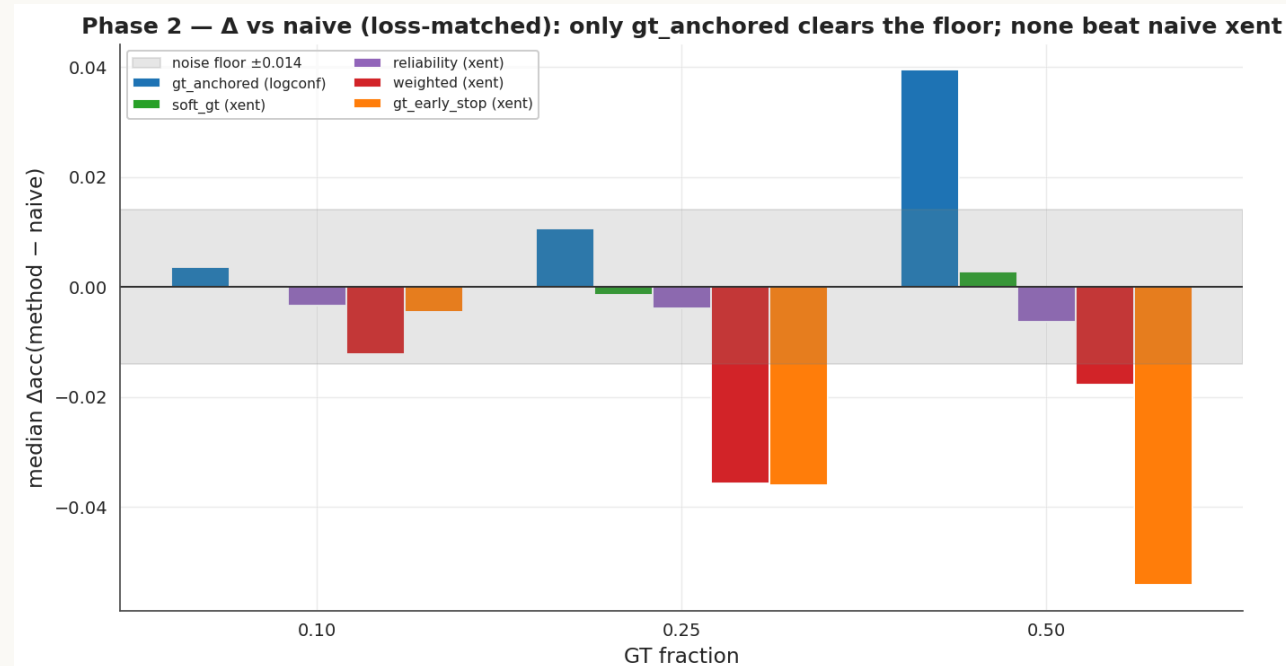
- An **oracle** uses held-out GT to place the budget exactly on the **teacher-wrong rows** — an upper bound on any allocation rule.
- Paired against random placement at matched budget: oracle - random = **+0.0006** (0.10), **-0.0049** (0.25) — both within the **MDE (0.0071)**, so a genuine null, not underpowered.
- If the best possible placement ties random, allocation heuristics have **little to gain** on this testbed.

Phase 1b · B — WHERE you spend GT does not matter (allocation NULL): a perfect error-targeting oracle ties random



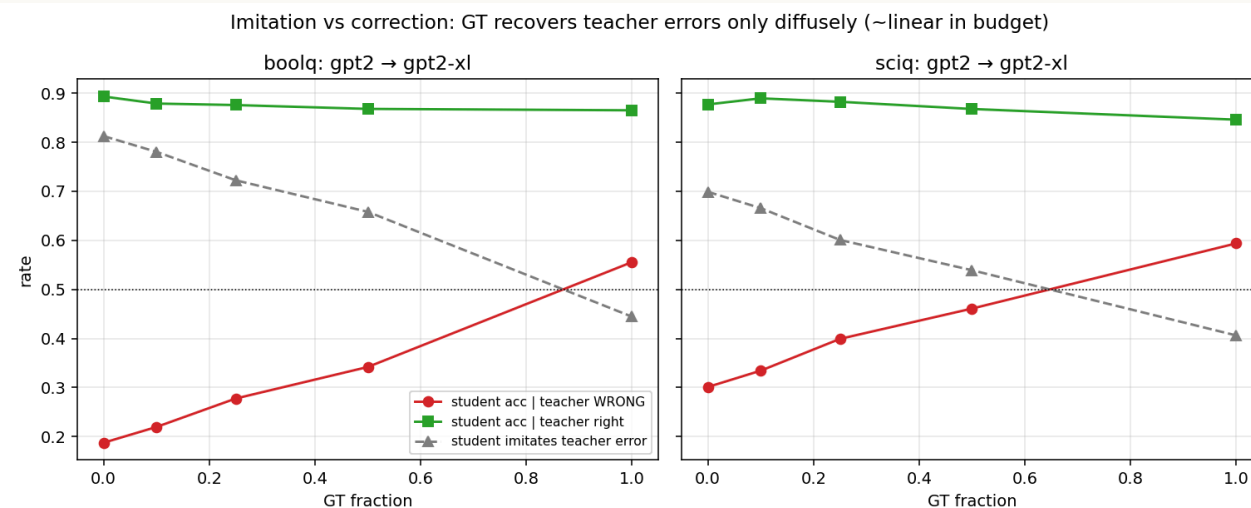
## How it's combined — within noise (combination null)

- Five methods vs naive mixing at matched budget: **M1** GT up-weighting · **M2** soft-GT targets · **M3** GT-anchored logconf · **M4** reliability-weighted weak labels · **M5** GT-based early stopping.
- Median  $\Delta$  vs naive  $\approx 0$  across  $\{0.10, 0.25, 0.50\}$ ; none shifts the curve left or raises the ceiling.
- **M3 (gt-anchored)** is the only method above the floor (+0.040 at 0.50), but it recovers logconf toward xent **without exceeding plain xent** (0.642 vs 0.697).



## Mechanism: the student reproduces the teacher's errors

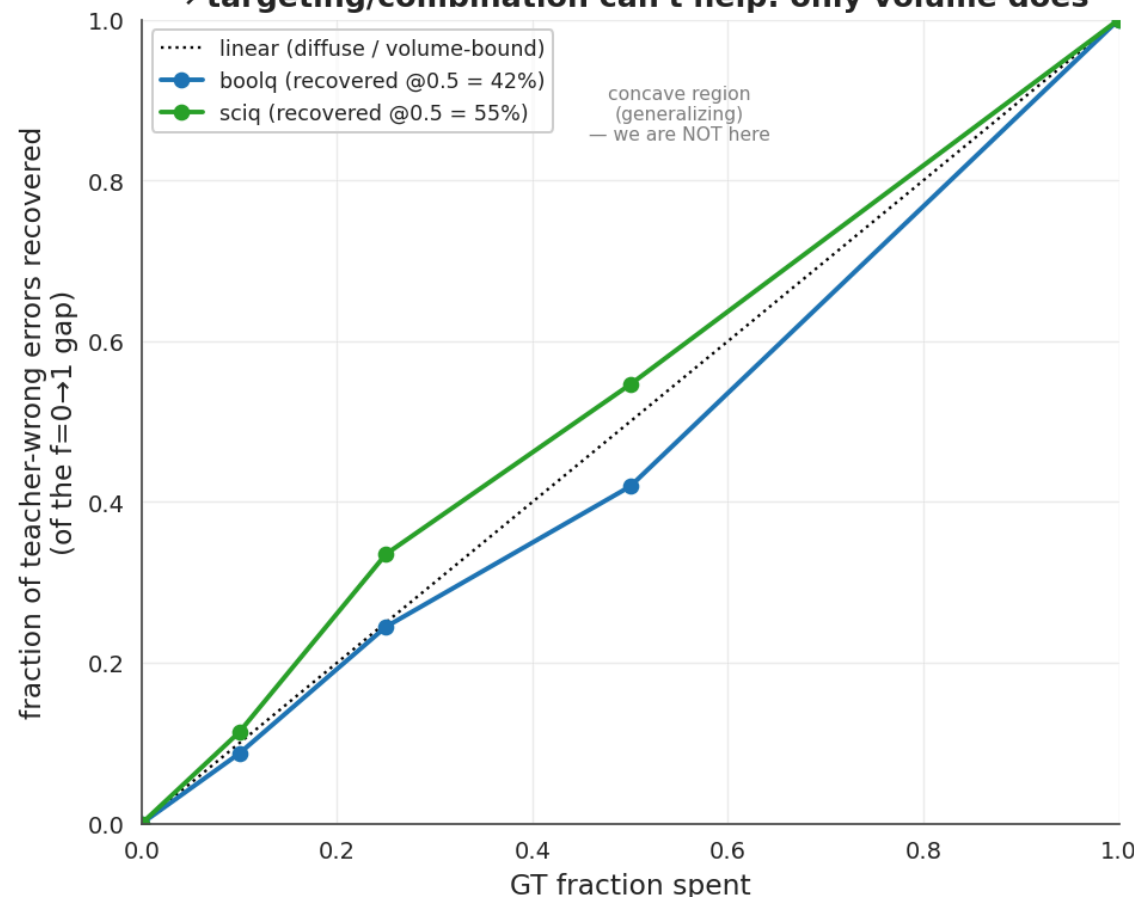
- Probe: gpt2 → gpt2-xl, joining teacher and student **per-example** test predictions.
- On teacher-wrong rows at 0% GT, the student reproduces the teacher's wrong answer **81% (BoolQ) / 70% (SciQ)** of the time.
- Errors are **largely inherited rather than independent** — so the rows needing correction are identifiable in principle.



## Mechanism: recovery is ~linear in budget (volume-bound)

- Recovery of teacher-wrong rows is **~linear in budget** — 42% / 55% of the 0→100% gap recovered at 50%, tracking  $y = x$  rather than a concave curve.
- A GT label's marginal value appears **roughly independent of which row it lands on**.
- Under volume-bound recovery, placement (*where*) and re-weighting (*how*) have **little leverage** — consistent with both nulls.

Mechanism — error recovery is ~LINEAR in budget (diffuse), not concave  
→ targeting/combination can't help: only volume does



# Pre-registration scorecard — hits and misses

---

**Phase 1** (P1–P6, git-anchored before seeds 0/2 existed)

|    | Prediction                        | Outcome                                      |
|----|-----------------------------------|----------------------------------------------|
| P1 | knee at ~25%                      | ✗ refuted — back-loaded, no knee (retracted) |
| P2 | $\leq 10\%$ GT inert              | ✓ within noise                               |
| P3 | mixing > GT-only                  | ✓ (confound named → resolved in 1b)          |
| P4 | logconf null                      | ✓ inert at every budget                      |
| P5 | scale interaction (gap → more GT) | — underpowered at GPT-2 scale                |
| P6 | 0.25–0.50 plateau                 | ✓ flat in that range                         |

**Phase 2** (M1–M5, registered before the portfolio ran): M2 ✓ null, M4 ✓ null (as flagged uncertain), **M3** 🟡 (strongest bet — rescues logconf, still < xent), M1/M5 ✗ negative.

## Experiments across the three axes

---

| Axis       | Experiment                                | Outcome                                    |
|------------|-------------------------------------------|--------------------------------------------|
| How much   | 8-point fraction sweep, $0 \rightarrow 1$ | back-loaded, saturates $\sim 0.75$         |
| Where      | error-targeting oracle + random control   | null – within MDE                          |
| How        | 5 combination / loss methods              | null – 1 above floor, still $< x_{ent}$    |
| Loss       | $x_{ent}$ vs logconf                      | logconf inert $\rightarrow$ dropped        |
| Tasks      | BoolQ + SciQ                              | replicates on both                         |
| Mechanism  | imitation-vs-correction probe             | recovery $\sim$ linear in budget           |
| Robustness | 8-seed variance; 5-seed baseline pass     | exclusion by rule; 3-seed conclusions hold |

Coverage spans the three axes plus loss, task, and robustness checks; the negatives are concordant with the volume-bound account.

## Takeaways

---

- Reproduced W2SG on the GPT-2 family and extended it to a supervision-budget setting.
- Decomposed the budget question into *how much* / *where* / *how* — a powered null on the last two, and a back-loaded, saturating curve on the first.
- One account — **inherited errors plus volume-bound recovery** — is consistent with all three.
- So the binding constraint here looks like **scale**, not allocation or combination — which sets the next experiments.

## Next steps

---

1. **Vary the capability gap.** Re-run the budget sweep with a bigger student-teacher gap (a weaker / handicapped teacher; larger families if allowed). If recovery turns from linear to **concave**, targeted GT starts to generalize — and *where* / *how* would matter again.
2. **Are the teacher's mistakes structured?** Train a small probe to predict, from the student's features, where the teacher is wrong. If it can't, the errors are scattered with no pattern to target — which is why allocation doesn't help. Cheap; reuses saved predictions.
3. **Schedule the budget over training.** We mixed GT and weak labels in a fixed ratio throughout, never varying the *timing*. Compare GT-first / annealed / interleaved — recent work suggests the schedule can matter more than the loss.
4. **Iterate the loop.** Use the budget-trained student as the next teacher and repeat at the same total GT. Does a budget that's null in one pass add up over rounds?

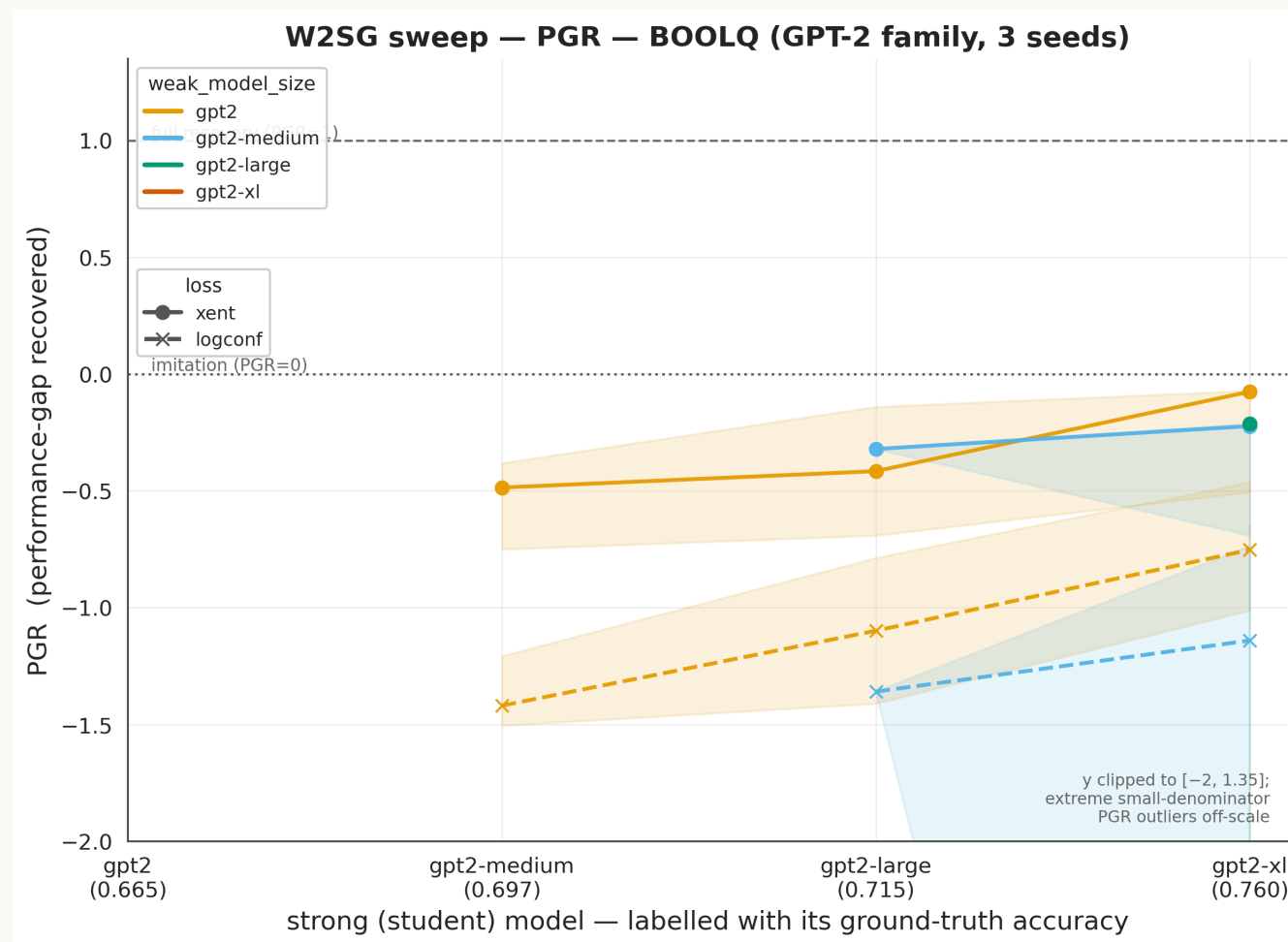
Ordered by relevance to the mechanism: (1) tests it directly · (2) explains it · (3-4) open new axes.

# Appendix

frac=0 reproduction (acc + PGR) · transfer heatmap · per-method × pair detail · de-confound · cross-task

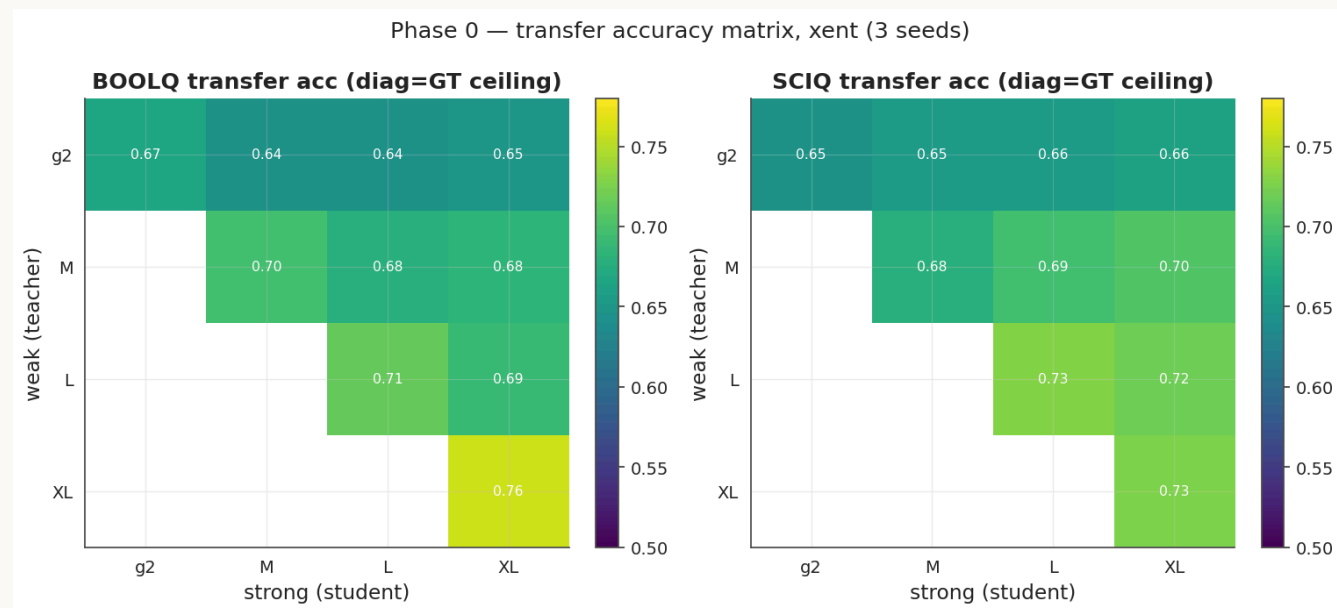
## Appendix — frac=0 reproduction

- Standard W2SG sweep, GPT-2 family, BoolQ, **0% GT**, PGR axis.
- Median xent PGR **-0.27**; several pairs fall below the imitation line — BoolQ is low-signal at 0% GT.
- This is the baseline the 25%-GT sweep is measured against.



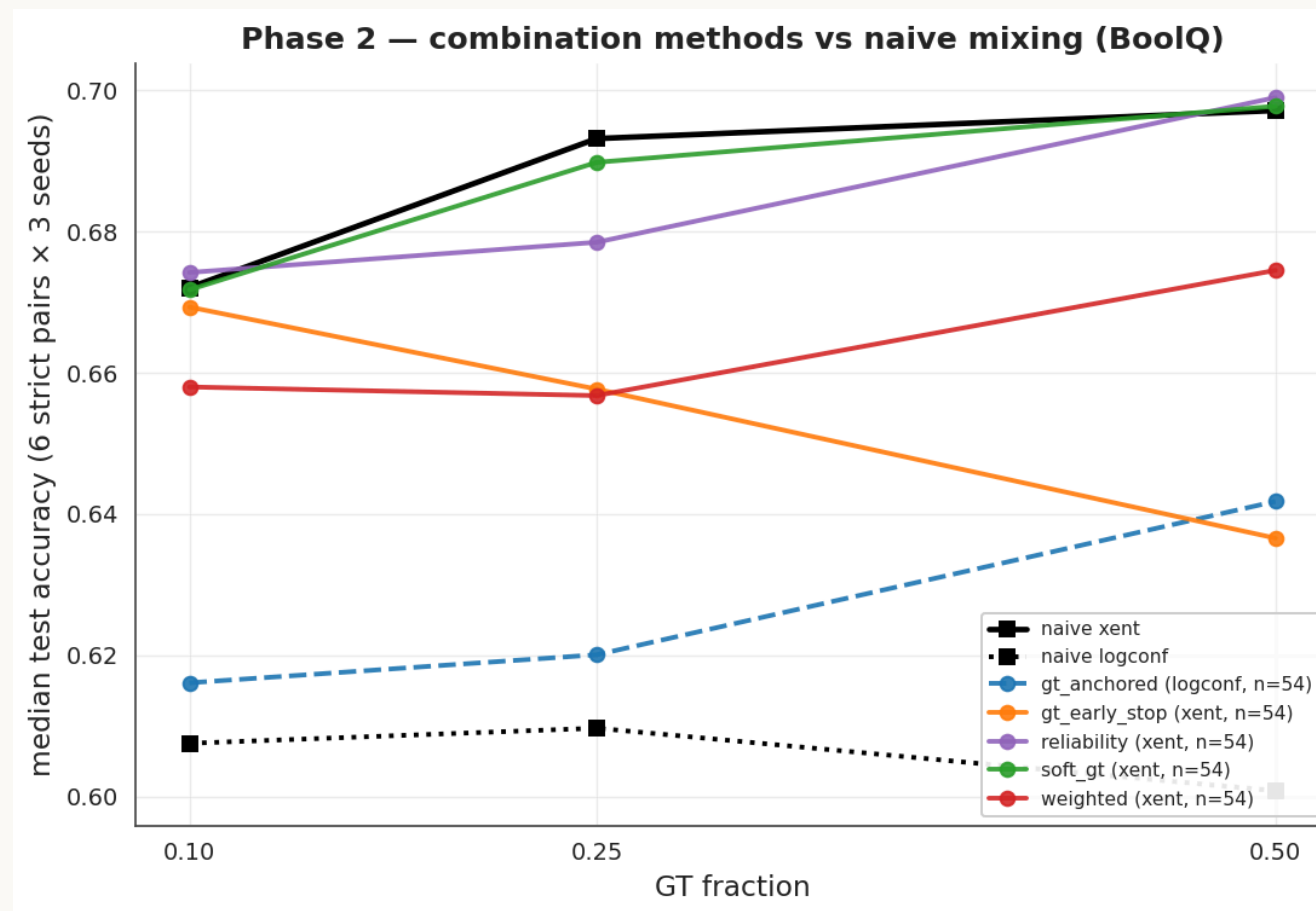
## Appendix — transfer tracks the teacher more than student capacity

- With the teacher fixed, scaling the student adds  $\sim +0.003$ ; with the student fixed, a better teacher adds  $\sim +0.04$ .
- Transfer tracks the teacher far more than student capacity — the static counterpart of the imitation result.



## Appendix – combination portfolio, per-method detail

- All five methods track naive mixing across  $\{0.10, 0.25, 0.50\}$ .
- gt-anchored (logconf) is the only method above the floor; it recovers logconf but stays under xent.



## Appendix — weak labels carry information (de-confound)

- Ordering: **random** < **gt\_only** < **naive mixing** (15/15 BoolQ, 18/18 SciQ pairs).
- Replacing weak labels with noise reduces accuracy → the mixing gain reflects weak-label information, not just added rows.

